



ITSIMPLERA INSTITUTE

Cybersecurity Analyst Internship

RED TEAM ASSIGNMENT # 03

NETWORK PROTOCOLS • WEB RECONNAISSANCE • ENUMERATION

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Task 1 : Protocols and Port Numbers

OBJECTIVE

The objective of this task is to study common network protocols, identify their default port numbers, determine whether they use TCP or UDP, understand their purpose, and identify whether they provide secure or insecure communication.

Table 1.1 — Common Network Protocols

Protocol	Default Port	TCP/UDP	Purpose	Secure / Insecure
HTTP	80	TCP	Used for transferring web pages between web servers and browsers.	Insecure
HTTPS	443	TCP	Provides secure communication between web servers and browsers using SSL/TLS encryption.	Secure
FTP	21	TCP	Used for transferring files between computers over a network.	Insecure
SFTP	22	TCP	Provides secure file transfer over SSH.	Secure
SSH	22	TCP	Allows secure remote access and administration of systems.	Secure
Telnet	23	TCP	Used for remote login without encryption.	Insecure
SMTP	25	TCP	Transfers outgoing email messages between mail servers.	Insecure
DNS	53	TCP/UDP	Resolves domain names into IP addresses.	Generally Insecure
DHCP	67 / 68	UDP	Automatically assigns IP addresses to devices on a network.	Insecure
POP3	110	TCP	Retrieves emails from a mail server to a client device.	Insecure
IMAP	143	TCP	Synchronizes emails between the mail server and client devices.	Insecure
SNMP	161	UDP	Monitors and manages network devices.	Insecure
LDAP	389	TCP/UDP	Provides directory services and authentication.	Insecure
SMB	445	TCP	Enables file and printer sharing over a network.	Insecure
RDP	3389	TCP	Allows users to remotely access Windows systems.	Secure (When Properly Configured)

Task 2 : HTTP vs HTTPS Analysis

OBJECTIVE

The objective of this task is to analyze the implementation of HTTP and HTTPS on different websites, determine whether they automatically redirect users from HTTP to HTTPS, and understand the importance of secure communication over the Internet.

PROCEDURE

1. Open a web browser such as Google Chrome or Mozilla Firefox.
2. Type `http://website.com` in the address bar.
3. Observe whether the website redirects to `https://website.com`.
4. Click on the padlock icon in the address bar to verify that the SSL/TLS certificate is valid.

Table 2.1 — HTTP vs HTTPS Analysis

Website	HTTP or HTTPS	Redirects HTTP→HTTPS	Secure?	Explanation
Google	HTTPS	Yes	Yes	Google automatically redirects HTTP requests to HTTPS. The SSL/TLS certificate encrypts communication between the browser and the server, protecting user data from interception.
GitHub	HTTPS	Yes	Yes	GitHub enforces HTTPS for all connections and uses a valid SSL/TLS certificate to ensure secure communication.
Facebook	HTTPS	Yes	Yes	Facebook redirects all HTTP traffic to HTTPS to protect user credentials and personal information.
YouTube	HTTPS	Yes	Yes	YouTube uses HTTPS to encrypt all transmitted data and prevent eavesdropping.
Wikipedia	HTTPS	Yes	Yes	Wikipedia provides encrypted communication using HTTPS and a valid SSL certificate.
Amazon	HTTPS	Yes	Yes	Amazon secures all user sessions with HTTPS to protect sensitive information such as login credentials and payment details.
LinkedIn	HTTPS	Yes	Yes	LinkedIn automatically redirects HTTP requests to HTTPS to provide secure communication.

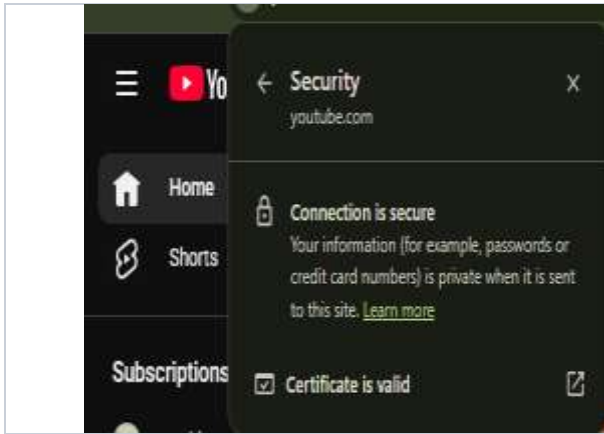


Figure 1: YouTube HTTPS Padlock Verification

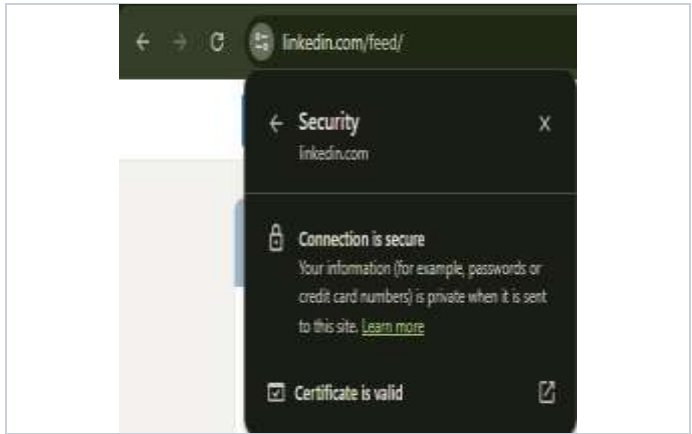


Figure 2: LinkedIn HTTPS Padlock Verification



OBSERVATION

All analyzed websites use HTTPS instead of HTTP. Each website automatically redirects users from HTTP to HTTPS and presents a valid SSL/TLS certificate. This ensures that the communication between the client and the server remains encrypted and protected from interception or modification.



CONCLUSION

HTTPS has become the standard protocol for modern websites because it provides confidentiality, integrity, and authentication through SSL/TLS encryption. Automatic redirection from HTTP to HTTPS helps prevent users from accidentally accessing insecure versions of websites and significantly improves web security.

Task 3 : Kali Linux Network Reconnaissance

OBJECTIVE

The objective of this task is to verify network connectivity and perform basic network reconnaissance using the **Ping** and **Nmap** tools in Kali Linux.

Part A: Network Connectivity using Ping

COMMAND

```
$ ping google.com
```

```
(kali㉿kali)-[~]  
$ ping google.com  
PING google.com (142.250.202.174) 56(84) bytes of data.  
64 bytes from lcmcta-ae-in-f14.1e100.net (142.250.202.174): icmp_se  
q=1 ttl=115 time=40.9 ms  
64 bytes from lcmcta-ae-in-f14.1e100.net (142.250.202.174): icmp_se  
q=2 ttl=115 time=38.5 ms  
64 bytes from lcmcta-ae-in-f14.1e100.net (142.250.202.174): icmp_se  
q=3 ttl=115 time=39.3 ms  
64 bytes from lcmcta-ae-in-f14.1e100.net (142.250.202.174): icmp_se  
q=4 ttl=115 time=38.7 ms  
64 bytes from lcmcta-ae-in-f14.1e100.net (142.250.202.174): icmp_se  
q=5 ttl=115 time=38.6 ms  
64 bytes from lcmcta-ae-in-f14.1e100.net (142.250.202.174): icmp_se  
q=6 ttl=115 time=39.5 ms  
64 bytes from lcmcta-ae-in-f14.1e100.net (142.250.202.174): icmp_se  
q=7 ttl=115 time=40.2 ms  
64 bytes from lcmcta-ae-in-f14.1e100.net (142.250.202.174): icmp_se  
q=8 ttl=115 time=36.8 ms  
64 bytes from lcmcta-ae-in-f14.1e100.net (142.250.202.174): icmp_se  
q=9 ttl=115 time=40.0 ms  
64 bytes from lcmcta-ae-in-f14.1e100.net (142.250.202.174): icmp_se  
q=10 ttl=115 time=38.3 ms  
64 bytes from lcmcta-ae-in-f14.1e100.net (142.250.202.174): icmp_se
```

Figure 3: Ping Command Output

This command checks whether the local system can communicate with **google.com** and measures the network response time.



OBSERVATION

- Target IP Address: **142.250.202.174**
- Average Response Time: **36.8–40.9 ms**
- TTL Value: **115**
- Result: All packets were received successfully, indicating stable network connectivity.

EXPLANATION

- **Response Time (ms):** Indicates the time taken for a packet to travel to the destination and back. Lower values indicate better network performance.
- **TTL (Time To Live):** TTL limits the number of network hops a packet can make before being discarded. A value of **115** indicates the packet successfully reached Google through several routers.

Part B: Find Local IP Address

COMMAND

```
$ ip a
```

```
(kali㉿kali)-[~]  
└─$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN  
   group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel s  
   tate UP group default qlen 1000  
    link/ether 08:00:27:d2:26:79 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixro  
   ute eth0  
        valid_lft 86003sec preferred_lft 86003sec  
    inet6 fe80::5056:de82:8157:619/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noque  
   ue state DOWN group default  
    link/ether 02:42:35:51:ff:1b brd ff:ff:ff:ff:ff:ff  
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0  
        valid_lft forever preferred_lft forever
```

Figure 4: Network Interface Configuration (ip a)

This command displays all network interfaces and their assigned IP addresses.

OBSERVATION

The active network interface is **eth0** with the IPv4 address **10.0.2.15/24**.

Part C: Nmap Scan of scanme.nmap.org

COMMAND

```
$ nmap scanme.nmap.org
```

```

(kali@kali)-[~]
$ nmap scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-18 20:11 +0500
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0075s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:9
1ff:fe18:bb2f
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
8600/tcp  closed asterix
16000/tcp closed fmsas

Nmap done: 1 IP address (1 host up) scanned in 20.29 seconds

```

Figure 5: Nmap Scan Output — scanme.nmap.org

This command scans the target host to identify open or closed TCP ports.

Port	State	Service
8600	Closed	Asterix
16000	Closed	fmsas

No open TCP ports were detected on the target during the scan.

Part D: Nmap Scan of Local System

COMMAND

```
$ nmap 10.0.2.15
```

```

(kali@kali)-[~]
$ nmap 10.0.2.15
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-18 20:13 +0500
Nmap scan report for 10.0.2.15
Host is up (0.000050s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 8.53 seconds

```

Figure 6: Nmap Scan Output — Local System (10.0.2.15)

This command scans the local machine to identify open ports and running services.

Port	State	Service	Purpose
22	Open	SSH	Secure remote login
80	Open	HTTP	Web server communication



POTENTIAL SECURITY RISK

Port 80 (HTTP) can be a potential security risk because it transmits data without encryption, making communication vulnerable to interception and man-in-the-middle attacks. A more secure alternative is HTTPS (Port 443).

NMAP -SV OPTION

The **-sv** option enables **service version detection**. It identifies the version of the services running on open ports, which helps security analysts detect outdated or vulnerable software.



CONCLUSION

The **Ping** command confirmed successful connectivity to Google with response times between **36.8 ms and 40.9 ms** and a **TTL value of 115**. The **Nmap** scan of **scanme.nmap.org** found no open ports, while the scan of the local machine identified **SSH (Port 22)** and **HTTP (Port 80)** as open services. These tools are useful for network reconnaissance and identifying potential security risks.

Task 4 : What Happens When You Type a URL and Press Enter?

OBJECTIVE

The objective of this task is to understand the complete process that takes place when a user enters a URL into a web browser. This includes DNS resolution, establishing a secure connection, requesting web resources, and rendering the webpage.

Step 1: User Enters a URL

The process begins when a user enters a URL (e.g., **https://www.google.com**) into the browser's address bar and presses **Enter**. The browser first checks whether the entered URL is valid.

Step 2: Browser Cache Check

The browser checks its local cache to determine whether the website's IP address or previously downloaded resources are already stored.

- If found, the browser uses the cached information.
- Otherwise, it proceeds with a DNS lookup.

Step 3: DNS Lookup

The browser needs the server's IP address because computers communicate using IP addresses rather than domain names.

The browser queries a DNS server to translate the domain name into its corresponding IP address.

Example: www.google.com → 142.250.202.174

Step 4: IP Address Resolution

After the DNS server returns the IP address, the browser now knows the location of the web server and can establish communication with it.

Step 5: TCP Three-Way Handshake

Before data can be exchanged, a TCP connection is established using the Three-Way Handshake.

STEP 1: SYN

The client sends a **SYN** packet to the server requesting a connection.

STEP 2: SYN-ACK

The server responds with a **SYN-ACK** packet, acknowledging the request.

STEP 3: ACK

The client sends an **ACK** packet, confirming the connection.

The TCP connection is now established.

Step 6: TLS/HTTPS Handshake

If the website uses **HTTPS**, a TLS handshake takes place before any data is exchanged. During this process:

- The server sends its SSL/TLS certificate.
- The browser verifies the certificate with a trusted Certificate Authority (CA).
- Encryption keys are exchanged securely.
- An encrypted communication channel is established.

This ensures the confidentiality, integrity, and authenticity of the communication.

Step 7: HTTP Request

Once the secure connection is established, the browser sends an HTTP request to the web server.

Example:

```
$ GET / HTTP/1.1
```

```
$ Host: www.google.com
```

The request asks the server to provide the requested webpage.

Step 8: Server Response

The web server processes the request and returns an HTTP response. The response contains:

- Status code (e.g., **200 OK**)
- HTML document
- CSS files
- JavaScript files
- Images and other resources

Step 9: Browser Renders the Webpage

The browser receives the response and begins rendering the webpage by:

1. Parsing the HTML.
2. Downloading CSS files.
3. Executing JavaScript.
4. Downloading images, fonts, and other resources.
5. Constructing the page layout.
6. Displaying the fully rendered webpage to the user.

✓ CONCLUSION

When a user enters a URL into a browser, several networking processes occur before the webpage is displayed. The browser resolves the domain name using DNS, establishes a TCP connection, performs a TLS handshake for secure communication, sends an HTTP request, receives the server's response, and finally renders the webpage. Understanding this process helps cybersecurity professionals analyze network communication and troubleshoot web-related issues.

Task 5 : Networking Cheat Sheet

OBJECTIVE

The objective of this task is to create a quick reference guide for common network protocols, including their default port numbers, transport protocols, purposes, security risks, and secure alternatives.

Networking Cheat Sheet — Reference Table

Protocol	Port	TCP/UDP	Purpose	Security Risk	Secure Alternative
HTTP	80	TCP	Transfers web pages between client and server.	Data is transmitted in plain text.	HTTPS
HTTPS	443	TCP	Provides secure web communication using SSL/TLS.	Misconfigured certificates may cause security issues.	N/A
FTP	21	TCP	Transfers files between systems.	Credentials and files are sent unencrypted.	SFTP / FTPS
SSH	22	TCP	Secure remote login and administration.	Weak passwords may allow brute-force attacks.	Use SSH keys and MFA
Telnet	23	TCP	Remote command-line access.	Sends all data in plain text.	SSH
SMTP	25	TCP	Sends email between mail servers.	Emails may be intercepted if not encrypted.	SMTPS
DNS	53	TCP/UDP	Resolves domain names into IP addresses.	Vulnerable to DNS spoofing and cache poisoning.	DNS over HTTPS (DoH) / DNS over TLS (DoT)
DHCP	67 / 68	UDP	Automatically assigns IP addresses to devices.	Rogue DHCP servers can assign malicious configurations.	DHCP Snooping
POP3	110	TCP	Downloads emails from a mail server.	Login credentials are sent unencrypted.	POP3S
IMAP	143	TCP	Synchronizes emails across multiple devices.	Unencrypted communication may expose data.	IMAPS
SNMP	161	UDP	Monitors and manages network devices.	Default community strings can expose network information.	SNMPv3
LDAP	389	TCP/UDP	Provides directory and authentication services.	Data may be transmitted without encryption.	LDAPS
SMB	445	TCP	File and printer sharing over networks.	Frequently targeted by malware and ransomware.	SMBv3 with encryption
RDP	3389	TCP	Remote access to Windows systems.	Vulnerable to brute-force attacks if exposed.	RDP with VPN, MFA, and NLA
NTP	123	UDP	Synchronizes system time across devices.	Can be abused for amplification attacks.	Authenticated NTP



SUMMARY

This networking cheat sheet provides a quick reference to commonly used network protocols, their default ports, purposes, associated security risks, and recommended secure alternatives. Using secure protocols such as HTTPS, SSH, SFTP, SNMPv3, and LDAPS helps protect data confidentiality, integrity, and authentication while reducing the risk of cyberattacks.

Task 6 : Kali Linux Web Enumeration

OBJECTIVE

The objective of this task is to gather publicly available information about websites using **WhatWeb** and **Whois**. These tools help identify web technologies and domain registration details.

Website 1: Google

COMMANDS

```
$ whatweb google.com
```

```
(kali㉿kali)-[~]  
└─$ whatweb google.com  
http://google.com [301 Moved Permanently] Country[UNITED STATES][US]  
, HTTPServer[gws], IP[142.250.202.174], RedirectLocation[http://www.  
google.com/], Title[301 Moved], UncommonHeaders[content-security-pol  
icy-report-only], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]  
http://www.google.com/ [200 OK] Cookies[AEC,NID,__Secure-STRP], Coun  
try[UNITED STATES][US], HTML5, HTTPServer[gws], HttpOnly[AEC,NID], I  
P[142.251.151.119], Script, Title[Google], UncommonHeaders[content-s  
ecurity-policy-report-only], X-Frame-Options[SAMEORIGIN], X-XSS-Pro  
tection[0]
```

Figure 7: WhatWeb Scan Output — google.com

```
$ whois google.com
```

```
(kali㉿kali)-[~]  
└─$ whois google.com  
Domain Name: GOOGLE.COM  
Registry Domain ID: 2138514_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.markmonitor.com  
Registrar URL: http://www.markmonitor.com  
Updated Date: 2019-09-09T15:39:04Z  
Creation Date: 1997-09-15T04:00:00Z  
Registry Expiry Date: 2028-09-14T04:00:00Z  
Registrar: MarkMonitor Inc.  
Registrar IANA ID: 292  
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com  
Registrar Abuse Contact Phone: +1.2086851750  
Domain Status: clientDeleteProhibited https://icann.org/epp#clien  
tDeleteProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#cli  
entTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clien  
tUpdateProhibited  
Domain Status: serverDeleteProhibited https://icann.org/epp#serve  
rDeleteProhibited  
Domain Status: serverTransferProhibited https://icann.org/epp#ser  
verTransferProhibited  
Domain Status: serverUpdateProhibited https://icann.org/epp#serve  
rUpdateProhibited  
Name Server: NS1.GOOGLE.COM  
Name Server: NS2.GOOGLE.COM
```

Figure 8: Whois Lookup Output — google.com

OBSERVATION

Web Technologies: HTML5, HTTP Server (gws), Cookies (AEC, NID, __Secure-STRP), JavaScript

Domain Registrar: MarkMonitor Inc.

Registration Date: 15 September 1997

Name Servers: ns1.google.com, ns2.google.com, ns3.google.com, ns4.google.com

Interesting Information: Google redirects HTTP to HTTPS and uses security headers such as X-Frame-Options and Content-Security-Policy.

Website 2: GitHub

COMMANDS

```
$ whatweb github.com
```

```
(kali@kali)-[~]
$ whatweb github.com
http://github.com [301 Moved Permanently] Country[UNITED STATES][US], IP[20.207.73.82], RedirectLocation[https://github.com/]
https://github.com/ [200 OK] Content-Language[en-US], Cookies[_gh_sess,_octo,logged_in], Country[UNITED STATES][US], Email[you@domain.com], HTML5, HTTPServer[github.com], HttpOnly[_gh_sess,logged_in], IP[20.207.73.82], Open-Graph-Protocol[object][1401488693436528], OpenSearch[/opensearch.xml], Script[application/json,module], Strict-Transport-Security[max-age=31536000; includeSubdomains; preload], Title[GitHub · Change is constant. GitHub keeps you ahead. · GitHub], UncommonHeaders[x-content-type-options,referer-policy,content-security-policy,x-github-request-id], X-Frame-Options[deny], X-XSS-Protection[0]
```

Figure 9: WhatWeb Scan Output — github.com

```
$ whois github.com
```

```
(kali@kali)-[~]
$ whois github.com
Domain Name: GITHUB.COM
Registry Domain ID: 1264983250_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-09-07T09:16:32Z
Creation Date: 2007-10-09T18:20:50Z
Registry Expiry Date: 2026-10-09T18:20:50Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: DNS1.P08.NSONE.NET
Name Server: DNS2.P08.NSONE.NET
Name Server: DNS3.P08.NSONE.NET
Name Server: DNS4.P08.NSONE.NET
Name Server: NS-1283.AWSDNS-32.ORG
Name Server: NS-1707.AWSDNS-21.CO.UK
Name Server: NS-421.AWSDNS-52.COM
```

Figure 10: Whois Lookup Output — github.com

OBSERVATION

Web Technologies: HTML5, HTTP Server (github.com), Cookies, Open Graph Protocol, Strict-Transport-Security (HSTS)

Domain Registrar: MarkMonitor Inc.

Registration Date: 09 October 2007

Name Servers: dns1.p08.nsonone.net, dns2.p08.nsonone.net, dns3.p08.nsonone.net, dns4.p08.nsonone.net, AWS DNS servers

Interesting Information: GitHub automatically redirects HTTP to HTTPS and enforces secure communication using HSTS.

CONCLUSION

The **WhatWeb** tool identified the web technologies and security headers used by Google and GitHub, while **Whois** provided domain registration details such as the registrar, creation date, and name servers. Both websites are registered with **MarkMonitor Inc.**, use HTTPS for secure communication, and implement security mechanisms to protect users and their data.

Overall Conclusion

This assignment strengthened my understanding of networking fundamentals and cybersecurity concepts through both theoretical research and practical exercises. I learned about common network protocols, secure web communication, DNS resolution, and the process of loading a website. I also gained hands-on experience with Kali Linux tools such as **Ping**, **Nmap**, **WhatWeb**, and **Whois**, which are commonly used for network reconnaissance and information gathering. These skills are essential for performing security assessments and understanding how modern networks and web applications operate.

References

- <https://nmap.org/docs.html>
- <https://github.com/urbanadventurer/WhatWeb>
- <https://lookup.icann.org>
- <https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/Overview>